

AHMAD SBAI

+34 678 680 253 | sbaia1870@gmail.com | Barcelona, Spain

PROFESSIONAL SUMMARY

Data Scientist and Software Engineer with a strong 3-year engineering foundation in Mechatronics. Driven by a natural focus on the software and data-architecture side of engineering problems, I executed a deliberate, strategic pivot into Data Science. Skilled in building modular Python pipelines, advanced statistical analysis workflows, and production-ready full-stack data applications. Currently completing a BSc in Data Science at Harbour.Space Barcelona while actively delivering real-world analytical tools.

WORK EXPERIENCE

Python Automation Developer (Contract)

ESTRACOM (Morocco) | 2024 – 2025

- Engineered a custom Digital Archiving System in Python, successfully digitizing the company's entire legacy paper infrastructure.
- Built a searchable relational database interface that reduced administrative document retrieval time by over 80%.

Freelance Software Developer

Remote | 2023 – Present

- Architected bespoke desktop applications for small business clients to automate inventory tracking and sales logging.
- Managed the full software development lifecycle from requirements gathering to deployment, reducing manual data entry errors by an estimated 40%.

Electronics & Programming Instructor (Intern)

Kidocode (Malaysia) | 2023

- Taught fundamental C/C++ and Python programming concepts by guiding students through the development of interactive robotics projects.

Head of Internal Affairs (Student Committee)

Harbour.Space | 2025 – Present

- Engineered "Your Voice, Right Now", a real-time, QR-based feedback and data-collection system to streamline communication across the student body.

FEATURED TECHNICAL PROJECTS

Data Analysis: Music Trends in the Streaming Era (1M+ Tracks)

- Engineered a Python pipeline to analyze a dataset of over 1M tracks, utilizing Welch's t-tests to statistically confirm a significant duration decline post-2018.
- Cleaned, preprocessed, and standardized high-variance audio datasets using IQR outlier detection and Z-score normalization.

HR Analytics Data Pipeline | Lead Developer

- Architected a modular, CRUD-based Python application utilizing structured data schemas for optimized employee data management.
- Engineered a predictive analytics engine to simulate, forecast, and visualize company-wide budget impacts based on salary adjustments.
- Designed a dynamic CustomTkinter graphical interface for real-time statistical aggregation and automated reporting.

Nexus: Full-Stack Data Service API

- Architected a high-performance backend using the Router-Service-Repository pattern, enforcing validation via Pydantic and adding secure API-Key authentication dependencies.
- Engineered relational schemas handling atomic transactions across 6 database entities and integrated asynchronous HTTPX REST API calls for live climate telemetry verification.
- Developed a multi-threaded admin control panel alongside a session-persistent Telegram Bot for non-blocking real-time CRUD manipulations.

The Numerical Showdown: Algorithmic Benchmarking Suite

- Implemented four root-finding algorithms (Newton-Raphson, Secant, Bisection, Fixed-Point) from mathematical first principles in Python, with zero external solver dependencies.
- Engineered a live benchmarking dashboard tracking microsecond execution times and convergence rates across algorithms, with a heuristic engine to recommend the optimal solver per function profile.

EDUCATION

Bachelor of Science in Data Science

Harbour.Space University (Barcelona, Spain) | 2025 – Present

Mechatronics Engineering Core (3 Years Completed)

Asia Pacific University | 2020 – 2024

- Completed foundational engineering coursework—including robotics, embedded systems, and automation control before executing a strategic career pivot into Data Science.

KEY SKILLS

- Data & Analytics:** Python (Pandas, NumPy, SciPy), SQL, Tableau, Excel, Predictive Modeling, NLP Sentiment Analysis
- Software & Architecture:** C/C++, MATLAB, Git/GitHub, Multi-threading, Modular Software Design, CustomTkinter (GUI), API Integration
- Engineering Concepts:** Real-Time Signal Processing, Autocorrelation, Process Automation

ADDITIONAL INFORMATION

- Portfolio:** [Ahmad Sbai](#)
- Github:** [unknownahmad](#)
- LinkedIn:** [ahmadsbai](#)
- Languages:** Arabic (Native), English (Full Professional), French (Intermediate)